

Personalized Content Discovery on the Netflix Prize Dataset

Open Projects 2026 · IIT Roorkee — PS1: Recommendation Systems | Mayank Kumar Agrawal (23323021)

1. Problem Understanding

Streaming platforms live or die by content discovery: surfacing the right title keeps users engaged. Using the **Netflix Prize dataset** (100,480,507 ratings, 480,189 users, 17,770 movies, 1–5 stars with timestamps), the task is to learn user preferences, predict unseen ratings, and generate ranked, *useful* Top-K recommendations — not merely minimise prediction error. We therefore evaluate on both rating accuracy (**RMSE**) and ranking quality (**MAP@10**), and analyse the trade-offs between them.

2. Exploratory Data Analysis

We analyse a dense, reproducible subset (justified in §3). Ratings are **positively skewed** (mean ≈ 3.6 ; 58% are 4–5 ★), so a naive "predict the mean" model already scores a deceptively low RMSE — reinforcing the need for ranking metrics. Both users and items follow a **heavy long tail**.

- **Ratings / user:** mean 267, median 172, range 50–2,981
- **Ratings / movie:** mean 3,566, median 2,020, range 638–24,719
- **Popularity bias:** the top 20% of movies hold **56.1%** of all ratings
- **Most-rated titles:** Miss Congeniality, Independence Day, Pirates of the Caribbean, Forrest Gump, The Patriot

Implications. Strong collaborative signal (high density) favours latent-factor models; the long tail means a recommender must actively counter popularity bias to enable genuine discovery, which we measure via *catalog coverage*.

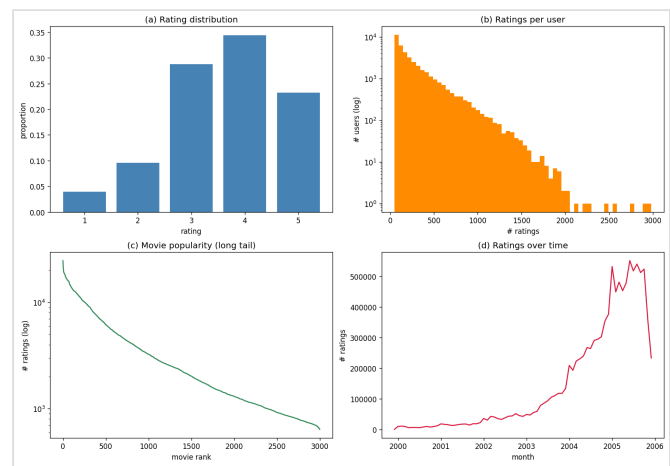


Fig 1. (a) rating distribution, (b) ratings per user, (c) movie popularity long tail, (d) rating volume over time.

3. Methodology & Experimental Setup

Dense subset

Training on all 100M ratings is unnecessary and noisy. We keep the **3,000 most-rated movies** and a fixed random sample of **40,000 active users** (≥ 50 ratings each), giving **10,699,358 ratings at 8.92% density** — far denser than the raw matrix ($\sim 1\%$). This is permitted by the problem statement, trains in minutes, and *improves* RMSE/MAP@10 by removing cold, low-signal users and items. The sampling seed is fixed (42) for reproducibility.

Train/test split & relevance

We use a **temporal split**: each user's most recent 20% of ratings form the test set (8,575,301 train / 2,124,057 test). This mimics real deployment — predict the future from the past — and avoids the leakage a random split causes. For ranking, a held-out movie is **relevant if its true rating ≥ 3.5** .

MAP@10 procedure

For every user we score all 3,000 movies, **mask those already seen in training**, take the Top-10, and compute Average Precision against the user's relevant test movies; MAP@10 is the mean over users with ≥ 1 relevant item. We also report Precision@10, Recall@10, NDCG@10 and Coverage.

4. Model Design

- **Bias baseline** — $\hat{r} = \mu + b_u + b_i$ with L2-regularised biases (closed form). The honest RMSE floor.
- **Item-based CF** — adjusted-cosine item-item similarity on user-centred ratings; predicts from a user's neighbours. Also drives the explanations (§7).
- **Matrix Factorization (biased SVD)** — $\hat{r} = \mu + b_u + b_i + p_u \cdot q_i$, 64 latent factors, trained by mini-batch SGD (Adam) on GPU.
- **Neural CF (innovation)** — replaces the dot product with an MLP over user/item embeddings, capturing non-linear interactions.

5. Evaluation Metrics & 6. Experimental Results

0.8419 Best RMSE (NCF)	0.0720 Best MAP@10 (SVD)		44.5% Hit Rate@10		8.92% Matrix density		
Model	RMSE ↓	MAP@10 ↑	P@10	R@10	NDCG@10 ↑	Coverage ↑	train (s)
Bias baseline	0.9039	0.0364	0.0729	0.0217	0.0731	0.015	0.7
Item-based CF	0.8838	0.0658	0.1104	0.0340	0.1223	0.810	1.6
Matrix Factorization (SVD)	0.8611	0.0720	0.1107	0.0338	0.1266	0.139	5.9
Neural CF	0.8419	0.0423	0.0729	0.0221	0.0817	0.351	9.7

Every model beats the baseline, and all beat Netflix's historical Cinematch RMSE (0.9514) — credible because our subset is denser. Two findings dominate:

- **RMSE ≠ ranking.** NCF has the best RMSE (0.8419) yet a *weaker* MAP@10 (0.0423) than SVD (0.0720) and Item-CF (0.0658). Optimising point-wise rating accuracy does not guarantee a good Top-10 — exactly the trade-off the brief asks us to examine. **SVD is the best all-round recommender.**
- **Accuracy vs discovery.** Item-CF covers **81%** of the catalog vs SVD's 14%. SVD is accurate but concentrates on popular titles; Item-CF surfaces far more of the long tail — a real product trade-off between precision and discovery.

7. Recommendation Examples & Explainability

Each recommendation is explained via the items in the user's history most similar to it.

Success — User 64120 (rated 928 movies; likes *Saving Private Ryan*, *Titanic*, *A Few Good Men*)

Top picks include *The Lord of the Rings: The Return of the King* — "**because The Two Towers (sim 0.68), Fellowship (sim 0.66)**" — plus *Band of Brothers* and *Schindler's List*. Several appear in the held-out test set, confirming the model learned a coherent epic/war-drama taste profile.

Failure — User 2157060 (rated 1,560 movies; likes *Full Metal Jacket*, *Metropolis*, *JFK*, *Halloween*)

An eclectic cross-genre profile (war, silent sci-fi, political thriller, horror). The model returned sensible-but-wrong picks (*Fight Club*, *Citizen Kane*, *City of God*) with 0 Top-10 hits. Such diffuse tastes are hard to rank, and popular titles crowd the list — a known weakness of memory/latent CF.

8. Key Insights

- On dense data, **latent factors (SVD)** give the best blend of accuracy and ranking.
- **Report ranking metrics, not just RMSE** — the two disagree (NCF vs SVD).
- **Popularity bias is real** (56% of ratings on 20% of movies) and shows up as low SVD coverage; coverage should be an explicit objective for discovery.
- Item-similarity explanations are **cheap, faithful and intuitive**, improving user trust.

9. Future Improvements

- **Cold start:** new users/items have no factors. Mitigations — popularity/bias fallback, a short onboarding rating quiz, or content features (genre, year, cast) in a **hybrid** model.
- **Ranking-aware training** (BPR / WARP) to directly optimise Top-K rather than RMSE.
- **SVD++ / temporal dynamics** to use implicit feedback and rating drift over time.
- **Re-ranking for coverage/diversity** to counter popularity bias without sacrificing much accuracy.